

History — Chemistry Chapter 6: Stoichiometry

The people behind the ideas. Every concept in this chapter was fought over, debated, and eventually figured out by real humans who made real mistakes. Here are their stories.

Topic 1 — Empirical vs Molecular Formula: The Birth of "What's in This Stuff?"

The alchemists' puzzle

Before chemistry was chemistry, it was **alchemy** — a mix of science, magic, and wishful thinking. For thousands of years, alchemists in Egypt, China, Persia, and Europe tried to turn lead into gold. They failed. But along the way, they discovered hundreds of new substances — acids, salts, metals, dyes — without having any clue what was inside them.

By the 1700s, people were tired of guessing. They started asking the question that would become modern chemistry: **what exactly is a compound made of?**

Joseph Proust and the law that started it all

In 1797, a French chemist named **Joseph Proust** did something revolutionary. He made the same compound — copper carbonate — in many different ways, from different starting materials. Every time he analyzed it, he found the same proportions of copper, carbon, and oxygen. No matter how he made it, it always had the SAME ratio of elements.

This was called the **Law of Definite Proportions**, and it was the birth of the idea that compounds have specific, fixed recipes. It's the reason we can write chemical formulas at all.

Without Proust's law, there would be no empirical formulas, because there would be no consistent ratios to describe. He was a quiet chemist, often overshadowed by his more famous rival Claude Berthollet, but Proust's insight is at the foundation of every formula you'll ever write.

Why "empirical"?

The word **empirical** comes from the Greek *empeirikos*, meaning "based on experience or experiment." Early chemists couldn't see molecules — they only had scales, flames, and patient observation. The empirical formula was literally the formula you could figure out from weighing things in a lab, without any assumptions about the inner structure of atoms. It was "chemistry from experience."

Later, as atomic theory developed, chemists realized the empirical formula was only PART of the story — it gave you ratios but not absolute counts. That's when the **molecular formula** was introduced, to capture the full picture of "how many atoms of each element are in one actual molecule."

Topic 2 — Molecular Mass: Weighing the Unseeable

Dalton's atoms

Until the early 1800s, the idea that matter was made of tiny indivisible particles — **atoms** — was mostly philosophical speculation, dating back to the ancient Greek thinker **Democritus** (around 400 BC). Democritus believed in atoms but had no evidence for them.

In 1803, an English schoolteacher named **John Dalton** changed everything. Dalton was a Quaker, a self-taught scientist, and colorblind (he described his own colorblindness in a paper — "Daltonism" is still a name for it). He proposed that every element had its own unique type of atom with a specific weight, and compounds formed when these atoms combined in whole-number ratios.

Dalton gave each element a weight **relative** to hydrogen (the lightest). Hydrogen = 1. Carbon = about 6 (he was a little off — correct is 12). Oxygen = about 8. This was the first **atomic mass table** in history.

Dalton couldn't weigh a single atom — nobody could, for over a century — but he figured out their *relative* masses from careful experiments on how much of one element combined with how much of another. From ratios alone, he unlocked the invisible.

The modern amu

Today we measure atomic mass in **atomic mass units (amu)**, also called **daltons** in Dalton's honor. One amu is defined as 1/12 the mass of a carbon-12 atom. That sounds arbitrary, but it's a careful choice — it makes every other element's atomic mass come out to a nice number close to a whole integer.

So when you calculate that water has a molecular mass of 18 amu, you're using a unit named after a Quaker schoolteacher from Manchester who first dreamed up the idea of atoms with specific weights.

Topic 3 — The Mole & Avogadro: Italy's Most Misunderstood Scientist

The tragic story of Amedeo Avogadro

Here's a name you'll never forget: **Lorenzo Romano Amedeo Carlo Avogadro, Count of Quaregna and Cerreto**.

That's his full name. He was an Italian nobleman born in 1776 in Turin, and he had every advantage — money, education, status. He became a lawyer, like his father. But chemistry and physics seduced him, and in 1811 he published a paper that should have made him instantly famous.

In that paper, Avogadro proposed what is now known as **Avogadro's Hypothesis**: *equal volumes of different gases, at the same temperature and pressure, contain equal numbers of molecules*.

This was a profound idea. It meant that if you had a balloon of hydrogen and a balloon of oxygen, both at the same conditions, they held the same number of molecules — even though the oxygen balloon would be heavier (because oxygen molecules are heavier).

But here's the tragedy: **almost no one believed him**.

Why? Because Avogadro was quiet, worked alone, and published in obscure Italian journals. The leading chemists of the time were in France, Germany, and England, and they mostly ignored him. For over **fifty years**, his brilliant idea was forgotten.

Avogadro died in 1856, having never received credit for his insight.

The rescue, four years late

In 1860, a chemistry conference was held in Karlsruhe, Germany, to try to settle a bunch of confusing arguments that had been slowing down chemistry for decades. A young Italian chemist named **Stanislao Cannizzaro** showed up with a pamphlet. In it, he reminded everyone about Avogadro's 50-year-old hypothesis and showed how, if you just accepted it, all the confusion dissolved — atomic masses, molecular formulas, everything suddenly made sense.

People listened. The pamphlet swept through the chemistry community like wildfire. Within a few years, Avogadro's hypothesis was accepted as a cornerstone of chemistry. The Italian nobleman who died in obscurity was posthumously hailed as a genius.

Today, in his honor, we call the number **6.022×10^{23} Avogadro's number**. It's the symbol **N_A**. The unit of "mole" was built around his insight.

Every time you see N_A in your homework, remember: it's named after a man whose greatest idea was ignored for most of his life.

Why 6.022×10^{23} ?

The specific value of Avogadro's number was measured much later, in the early 1900s, by scientists including **Jean Perrin** (who won the 1926 Nobel Prize for his measurements of Avogadro's number via Brownian motion). The modern value is $6.02214076 \times 10^{23}$.

In 2019, the international committee that sets measurement standards officially **redefined** the mole to be exactly $6.02214076 \times 10^{23}$ particles. Before 2019, the mole was defined based on 12 grams of carbon-12; now, Avogadro's number itself is the definition. This shows how central the idea has become.

Topic 4 — Molar Mass: The Bridge Engineered Over Centuries

There's no single "inventor" of molar mass — it emerged gradually from the work of many chemists as they realized that atomic masses in amu and molar masses in grams were numerically identical. That's not a coincidence; it's the brilliant design of the mole itself.

But a few figures stand out.

Jean-Baptiste Dumas — the meticulous measurer

In the 1820s and 30s, a French chemist named **Jean-Baptiste Dumas** perfected techniques for measuring the molar masses of gases and vapors. His methods were so precise that his atomic mass values held up for decades. Dumas also trained some of the next generation's greatest chemists — including Louis Pasteur.

Friedrich Ostwald — the fighter for the mole

At the end of the 19th century, a German chemist named **Friedrich Wilhelm Ostwald** was one of the first to strongly

champion the concept of the mole as a unit of measurement. Ironically, Ostwald didn't even believe atoms were real at first — he thought they were just useful mental shortcuts. Even so, he pushed for the mole to become a standard unit because it worked, regardless of whether atoms actually existed.

He eventually came around on atoms, after Einstein explained Brownian motion in 1905 in a way that only made sense if atoms were real. Ostwald won the Nobel Prize in Chemistry in 1909 — for his work on catalysis, not the mole, but he's a fun character in this story.

Topic 5 — Ionic Compounds: From Salt to Electrons

The story of electricity and chemistry

In 1800, an Italian scientist named **Alessandro Volta** invented the first battery (the "voltaic pile"). Suddenly, scientists could run controlled electric currents through stuff — and they started running currents through chemicals to see what happened.

In 1807, a brilliant, ambitious English chemist named **Humphry Davy** used a powerful battery to do something no one had ever done: he pulled pure sodium metal out of molten caustic soda (NaOH), and pure potassium out of molten potash (KOH). These were the first times in history anyone had isolated these elements.

Davy realized that ionic compounds are held together by **electric forces** — positive ions attracted to negative ions. That's still how we describe them today.

Svante Arrhenius — the kid who figured out ions

In 1884, a young Swedish graduate student named **Svante Arrhenius** submitted his PhD thesis. In it, he argued that when you dissolve a compound like NaCl in water, it actually splits into positive and negative ions (Na⁺ and Cl⁻) that float around independently.

His professors **hated** the idea. It sounded crazy. They gave him the lowest possible passing grade for his thesis — barely a pass.

Eighteen years later, in 1903, Arrhenius won the **Nobel Prize in Chemistry** for that very same thesis. The examiners were embarrassed. Arrhenius's idea of ionic dissociation became a cornerstone of chemistry, and it's why today we can write:



Every time you split an ionic compound into ions in a net ionic equation, you're using Arrhenius's big idea — the one his professors almost failed him for.

Topic 6 — Balancing Equations: The Lavoisier Revolution

The man who changed chemistry — and paid for it

Antoine-Laurent de Lavoisier is often called the "father of modern chemistry." He was born in Paris in 1743, into a wealthy family. He married a brilliant woman, **Marie-Anne Pierrette Paulze**, when he was 28 and she was 13 (yes, that was normal at the time — she became his scientific collaborator, translator, and illustrator).

Lavoisier worked meticulously. In the 1770s and 80s, he performed experiment after experiment showing that when things burn, they don't lose "phlogiston" (a fake substance that chemists thought escaped from burning objects). Instead, he showed that burning materials **combine with oxygen** from the air — and the total mass of everything involved stays exactly the same.

This was the **Law of Conservation of Mass**: matter is neither created nor destroyed in a chemical reaction.

This law is the entire reason you balance chemical equations. If Lavoisier hadn't nailed this down, there would be no reason to make sure the atoms match on both sides.

The guillotine

In 1789, the French Revolution broke out. Lavoisier, despite his science, had been a tax collector for the French king (he took the job to fund his research). When the revolutionaries came for the aristocracy, they came for Lavoisier.

In 1794, at age 50, Lavoisier was tried, convicted, and **beheaded by guillotine**. The judge reportedly said: "The Republic has no need of savants."

His friend, the mathematician **Joseph-Louis Lagrange**, said the next morning: *"It took them only an instant to cut off his head, but France may not produce another such head in a century."*

Lavoisier's notebooks, his brilliant wife, and his unfinished work survived. The Law of Conservation of Mass survived him. Every time you balance an equation, you're honoring a man whose insights outlived his very unjust execution.

Marie-Anne Lavoisier — the forgotten half of the revolution

Marie-Anne was more than Lavoisier's assistant. She learned English and translated key foreign chemistry works so Lavoisier could read them. She illustrated his experiments with precision. After his execution, she spent years trying to preserve his legacy, eventually publishing his memoirs.

She was one of chemistry's first great female scientists, working at a time when women weren't allowed into universities. Her story is less famous than her husband's but just as important.

Topic 7 — Ionic Equations: Following Arrhenius's Thread

The idea of ionic equations flows directly from Svante Arrhenius's theory of ionic dissociation (see Topic 5).

Once chemists accepted that dissolved salts exist as free ions, it became natural to ask: "In a reaction, which ions are actually doing something, and which are just floating around?" The concept of **spectator ions** — ions that don't participate in the reaction — emerged in the early 20th century as chemists tried to write cleaner, more informative equations.

The net ionic equation became the "stripped-down" version of a reaction, showing only the real chemical change. It's like the difference between a news story that includes every irrelevant detail and one that focuses on what actually happened.

By the 1930s, ionic equations were standard in chemistry textbooks. Today they're a fundamental tool, but their roots go back to that angry thesis committee in Sweden that almost failed Arrhenius.

Topic 8 — Structural Formulas: Drawing the Invisible

The dream of the benzene ring

Here's one of chemistry's most famous stories.

In the mid-1800s, chemists knew the molecular formula of benzene was C_6H_6 , but nobody could figure out how the atoms were arranged. The usual rules of bonding didn't seem to work — there were too many hydrogen atoms for a normal chain, and too few for anything else that made sense.

In 1865, a German chemist named **Friedrich August Kekulé** was dozing in front of a fireplace. He later claimed (the story may be partly mythic) that he saw a vision of a snake biting its own tail — an **ouroboros** from ancient Egyptian mythology. He woke up with a sudden idea: what if the six carbon atoms in benzene formed a **ring**, each bonded to one hydrogen?

This solved the puzzle. Benzene wasn't a chain; it was a hexagonal ring.

Kekulé's ring structure became one of the most iconic drawings in all of chemistry. It's still on the first page of every organic chemistry textbook. And it marks the moment when chemists realized that **drawing molecules** — not just writing formulas — could unlock deep truths about how they behaved.

Wöhler, urea, and the birth of organic chemistry

In 1828, another German chemist, **Friedrich Wöhler**, did something that shocked the scientific world. He took a simple inorganic compound (ammonium cyanate, NH_4OCN) and by accident heated it into something that tasted and smelled exactly like **urea** ($CO(NH_2)_2$) — the compound found in urine.

Why was this shocking? Because everyone believed "organic" compounds (those found in living things) could only be made by living things — a mysterious force called the **vital force**. Wöhler had just made an organic compound from a non-organic one without any life involved.

He wrote to his friend **Jöns Jacob Berzelius**: *"I must tell you that I can prepare urea without requiring a kidney of an animal, either man or dog."*

This was the start of organic chemistry as a real field, and it led directly to the development of structural formulas. Chemists realized that to understand compounds like urea and glucose, they needed to know not just which atoms were present, but HOW they were connected.

Structural formulas were born from the ashes of the "vital force" idea.

One last note — why history matters

You might be thinking: "Why do I need to know about a French guy who got beheaded or an Italian count who was ignored?"

Because science isn't a stack of facts — it's a conversation that's been going on for thousands of years. The people in this chapter were confused, excited, wrong, right, ambitious, persecuted, rich, poor, famous, ignored. Some died for their ideas. Some were robbed of credit. Some were lucky enough to see their work celebrated in their lifetime.

When you learn the mole concept, you're joining a conversation that started with Avogadro in 1811, went ignored for fifty years,

got rescued by Cannizzaro, nailed down by Perrin, and refined again in 2019 by international committees. You're part of that chain now.

That's not just history. That's what it means to be a scientist.